

INTRODUCTION TO APPLIED NUMERICAL METHODS

Curve Fitting: General Form of Newton's Interpolating Polynomials

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NTH-ORDER POLYNOMIAL INTERPOLATION

nth-Order Polynomial Interpolation

$$f_n(x) = b_0 + b_1(x - x_0) + \cdots + b_n(x - x_0)(x - x_1) \cdots (x - x_{n-1})$$

- (n+1) data points are required

$$[x_0, f(x_0)], [x_1, f(x_1)], \dots, [x_n, f(x_n)]$$



Coefficients to be evaluated:

$f[]$: finite divided difference

$$b_0 = f(x_0)$$

$$b_1 = f[x_1, x_0]$$

$$b_2 = f[x_2, x_1, x_0]$$

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$$b_n = f[x_n, x_{n-1}, \dots, x_1, x_0]$$

$$f[x_i, x_j] = \frac{f(x_i) - f(x_j)}{x_i - x_j}$$

First finite divided difference

$$f[x_i, x_j, x_k] = \frac{f[x_i, x_j] - f[x_j, x_k]}{x_i - x_k}$$

Second finite divided difference

$$f[x_n, x_{n-1}, \dots, x_1, x_0] = \frac{f[x_n, x_{n-1}, \dots, x_1] - f[x_{n-1}, x_{n-2}, \dots, x_0]}{x_n - x_0}$$

nth finite divided difference

NTH-ORDER POLYNOMIAL INTERPOLATION

Finite Divided Difference:

$$f[x_i, x_j] = \frac{f(x_i) - f(x_j)}{x_i - x_j}$$

First finite divided difference

$$f[x_i, x_j, x_k] = \frac{f[x_i, x_j] - f[x_j, x_k]}{x_i - x_k}$$

Second finite divided difference

$$f[x_n, x_{n-1}, \dots, x_1, x_0] = \frac{f[x_n, x_{n-1}, \dots, x_1] - f[x_{n-1}, x_{n-2}, \dots, x_0]}{x_n - x_0}$$

nth finite divided difference

recursive

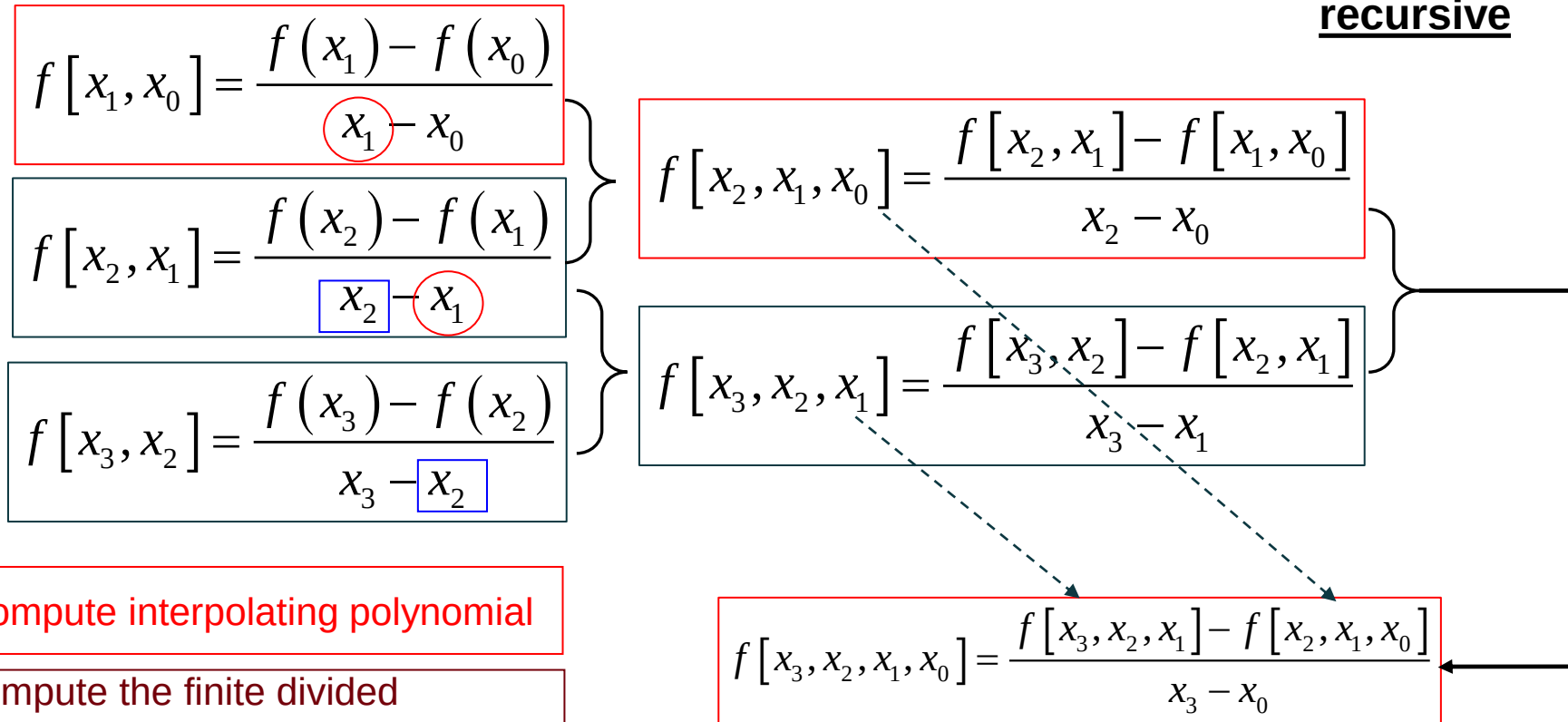
Higher order differences are computed by taking differences of lower-order differences

<i>i</i>	x_i	$f(x_i)$	First	Second	Third
0	x_0	$f(x_0)$	$f[x_1, x_0]$	$f[x_2, x_1, x_0]$	$f[x_3, x_2, x_1, x_0]$
1	x_1	$f(x_1)$	$f[x_2, x_1]$	$f[x_3, x_2, x_1]$	
2	x_2	$f(x_2)$	$f[x_3, x_2]$		
3	x_3	$f(x_3)$			

NTH-ORDER POLYNOMIAL INTERPOLATION

Finite Divided Difference (Detailed Steps):

recursive



NTH-ORDER POLYNOMIAL INTERPOLATION

nth-Order Polynomial Interpolation

$$f_n(x) = b_0 + b_1(x - x_0) + \cdots + b_n(x - x_0)(x - x_1) \cdots (x - x_{n-1})$$

$$b_0 = f(x_0)$$

$$b_1 = f[x_1, x_0]$$

$$b_2 = f[x_2, x_1, x_0]$$

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$$b_n = f[x_n, x_{n-1}, \dots, x_1, x_0]$$



$$f_n(x) = f(x_0) + (x - x_0)f[x_1, x_0] + (x - x_0)(x - x_1)f[x_2, x_1, x_0] \\ + \cdots + (x - x_0)(x - x_1) \cdots (x - x_{n-1})f[x_n, x_{n-1}, \dots, x_0]$$

Newton's divided-difference interpolating polynomial

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- Data points do not need to be equally spaced
 - The abscissa values do not need to be in ascending order

AN EXAMPLE

Estimate the natural logarithm of $x = 2$ using a third-order Newton's interpolation polynomial
(True value of $\ln(2) = 0.6931472$)

$$x_0 = 1 \quad f(x_0) = 0$$

$$x_1 = 4 \quad f(x_1) = 1.386294$$

$$x_2 = 6 \quad f(x_2) = 1.791759$$

$$x_3 = 5 \quad f(x_3) = 1.609438$$

Solution:

$$f_3(x) = f(x_0) + (x - x_0)f[x_1, x_0] + (x - x_0)(x - x_1)f[x_2, x_1, x_0] + (x - x_0)(x - x_1)(x - x_2)f[x_3, x_2, x_1, x_0]$$

First Finite Divided Difference:

$$f[x_1, x_0] = \frac{1.386294 - 0}{4 - 1} = 0.4620981$$

$$f[x_2, x_1] = \frac{1.791759 - 1.386294}{6 - 4} = 0.2027326$$

$$f[x_3, x_2] = \frac{1.609438 - 1.791759}{5 - 6} = 0.1823216$$

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Estimate the natural logarithm of $x = 2$ using a third-order Newton's interpolation polynomial
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Solution:

$$f_3(x) = f(x_0) + (x - x_0)f[x_1, x_0] + (x - x_0)(x - x_1)f[x_2, x_1, x_0] + (x - x_0)(x - x_1)(x - x_2)f[x_3, x_2, x_1, x_0]$$

Second Finite Divided Difference:

$$f[x_1, x_0] = \frac{1.386294 - 0}{4 - 1} = 0.4620981$$

$$f[x_2, x_1] = \frac{1.791759 - 1.386294}{6 - 4} = 0.2027326$$

$$f[x_3, x_2] = \frac{1.609438 - 1.791759}{5 - 6} = 0.1823216$$

$$f[x_2, x_1, x_0] = \frac{0.2027326 - 0.4620981}{6 - 1} = -0.05187311$$

$$f[x_3, x_2, x_1] = \frac{0.1823216 - 0.2027326}{5 - 4} = -0.02041100$$

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Estimate the natural logarithm of $x = 2$ using a third-order Newton's interpolation polynomial
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Solution:

$$f_3(x) = f(x_0) + (x - x_0) f[x_1, x_0] + (x - x_0)(x - x_1) f[x_2, x_1, x_0] + (x - x_0)(x - x_1)(x - x_2) f[x_3, x_2, x_1, x_0]$$

Third Finite Divided Difference:

$$f[x_2, x_1, x_0] = \frac{0.2027326 - 0.4620981}{6 - 1} = -0.05187311$$

$$f[x_3, x_2, x_1] = \frac{0.1823216 - 0.2027326}{5 - 4} = -0.02041100$$

$$f[x_3, x_2, x_1, x_0] = \frac{-0.02041100 - (-0.05187311)}{5 - 1} = 0.007865529$$

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Estimate the natural logarithm of $x = 2$ using a third-order Newton's interpolation polynomial
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Solution:

$$f_3(x) = f(x_0) + (x - x_0)f[x_1, x_0] + (x - x_0)(x - x_1)f[x_2, x_1, x_0] + (x - x_0)(x - x_1)(x - x_2)f[x_3, x_2, x_1, x_0]$$

$$b_0 = f(x_0) \quad b_1 = f[x_1, x_0] = 0.4620981 \quad b_2 = f[x_2, x_1, x_0] = -0.05187311$$

$$b_3 = f[x_2, x_2, x_1, x_0] = -0.05187311$$



$$f_3(x) = 0 + 0.4620981(x - 1) - 0.05187311(x - 1)(x - 4) + 0.007865529(x - 1)(x - 4)(x - 6)$$



$$f_3(2) = 0.6287686$$

$$\varepsilon_t = \left| \frac{f_t - f_1}{f_t} \right| \times 100\% = \left| \frac{0.6931472 - 0.6287686}{0.6931472} \right| \times 100\% = 9.3\%$$